MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 5C22

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, hydrogen

See figure 1

Mounting position: Any

Weight: 12-ounces nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epy	epx	Ebb	Ecc	eqx	egy	ib	Ib
Unit:	V	kv	kv	Vdc	Vdc	v	v	a	mAdc
Maximum:	6.8	16.0	16.0	---	---	200	---	325	200
Minimum:	5.8	--- (See note 1)	5% epy (See note 2)	4,500	---	---	(See note 3)	---	---
TEST CONDITIONS:	---	16.0	---	---	0	---	150	---	---

ABSOLUTE RATINGS:

Parameter:	tk	dik/dt	Pb	TA	prr
Unit:	sec	a/µs	----	°C	---
Maximum:	---	1,500	3.2×10^9	90	---
Minimum:	300	---	----	-50	---
TEST CONDITIONS:	---	-----	------	---	1,000

GENERAL:

Qualification - Required

5C22
FSC 5960

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						Min	Max	
	Quality conformance inspection, part 1							
3241	Heater current	Ef = 6.3 Vac	0.65	II	If	9.6	11.6	Aac
3247	Instantaneous starting	Ef = 6.8 Vac; epy = 13.5 kv (min); tk = 300 See notes 6 and 8	0.65	II	---	---	---	---
3246	Operation (1)	Ef = 5.8 Vac; epy = 18.0 kv (min); prr = 800; t = 300 See notes 6 and 7	0.65	II	egy	---	150	v
3247	DC anode voltage for conduction	See note 6	0.65	II	Ebb	---	2,500	Vdc
3251	Pulse emission	Ef = 6.3 Vac; ik = 325 a (min); prr = 60 ±10%; tp = 5 μs ±10%; starting time of current pulse = 2.5 μs (max); specified time interval = 4.0 μs tr = 0.5 μs (max)	0.65	II	egk	---	175	v
	Quality conformance inspection, part 2							
Appendix D, 30(b)	Dimensions	See figure 1	----	--	---	---	---	---
1071	High frequency vibration	No voltages applied	----	--	---	---	---	---
3246	Operation (1A)	Operation (1); Ef = 6.8 Vac	----	--	egy	---	150	v
3256	Anode delay time	Operation (1); t = 120; egy = 150 v	----	--	tad	---	0.65	μs
3256	Anode delay time drift	Anode delay time See note 9	----	--	Δtad	---	0.10	μs
3261	Time jitter	Ef = 6.3 Vac; epy = 5.0 kv (max) See note 6	----	--	tj	---	0.005	μs

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3					
----	Life-test	Group C; Ef = 6.3 Vac; t = 500 hrs.; 96 hrs. "on" and 1 hr. "off"; tube mounted in horizontal position See note 6	---	---	---	---
	Life-test end points:					
3246	Operation (1) and (1A)	See note 10	egy	---	160	v
3247	DC anode voltage for conduction	egy = 160 v	Ebb	---	4,000	Vdc
3261	Time jitter	egy = 160 v	tj	---	0.01	µs
3256	Anode delay time	egy = 160 v	tad	---	0.7	µs
3256	Anode delay time drift	egy = 160 v	Δ tad	---	0.1	µs
Appendix D, 30(b)	Dimensions	See figure 1 and note 11	---	---	---	---
1031	Variable-frequency vibration	No voltages applied See notes 4 and 11	---	---	---	---
3246	Operation (2)	TA = 90°C; t = 5.0 hours; Ef = 6.3 +0.5 Vac See notes 6, 7 and 11	egy	---	150	v
1041	Shock	Angle = 13° See notes 5 and 11	---	---	---	---
----	Shock end points:					
3246	Operation (1)		egy	---	150	v
3247	DC anode voltage for conduction		Ebb	---	2,500	Vdc
3261	Time jitter		tj	---	0.005	µs

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NOTES:

1. For instantaneous starting applications where anode voltage is applied instantaneously the maximum permissible epy is 13.5 kv and shall not be attained in less than 0.04 second.

2. In pulsed operation, the peak inverse voltage, exclusive of a spike of 0.05 µs maximum duration, shall not exceed 5.0 kv during the first 25 µs after the pulse.

3. Driver pulse measured at tube socket with thyratron grid disconnected shall be ecy = 200 v minimum; time of rise = 0.5 µs maximum; grid pulse duration = 2 µs minimum; impedance of driver circuit = 500 ohms maximum.

4. There shall be no pronounced resonance in the specified frequency range.

5. Use clamp as specified on Drawing 243-JAN.

6. The anode circuit constants shall be chosen so that at epy = 16.0 under resonant charging conditions:

   dik/dt = 1,500 a/µs minimum; ib = 175 a; tp = 1.0 µs ±10 percent; prr = 1,000.

   The grid pulse characteristic shall be tp = 2.0 µs maximum; tr = 0.5 µs minimum. Internal impedance of driver = 500 ohms minimum.

7. The tube shall operate continuously without evidence of arc-back.

8. Pushbutton starting shall be attempted up to three times, with intervals of from 10 to 30 seconds between successive attempts. Any tube failing to operate satisfactorily when the anode voltage (epy) applied rises from 0 to 13.5 kv within 0.03 second (the filter in the rectifier shall be designed so that the epy reaches at least 7 kv within 0.015 second) will be considered a failure.

9. During the interval between 2 minutes and 7 minutes of the anode delay time test, the change in anode delay time (Δtad) relative to the tad value observed on the anode delay time test shall not exceed the specified value.

10. Anode heating shall not be cause for rejection on operation (1) and operation (1A) performed during life-test end point test.

11. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of Quality Conformance Inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

12. Reclaimed materials shall be utilized to the maximum extent possible.

Custodians:
Army - EL
Navy - EC
Air Force - 85

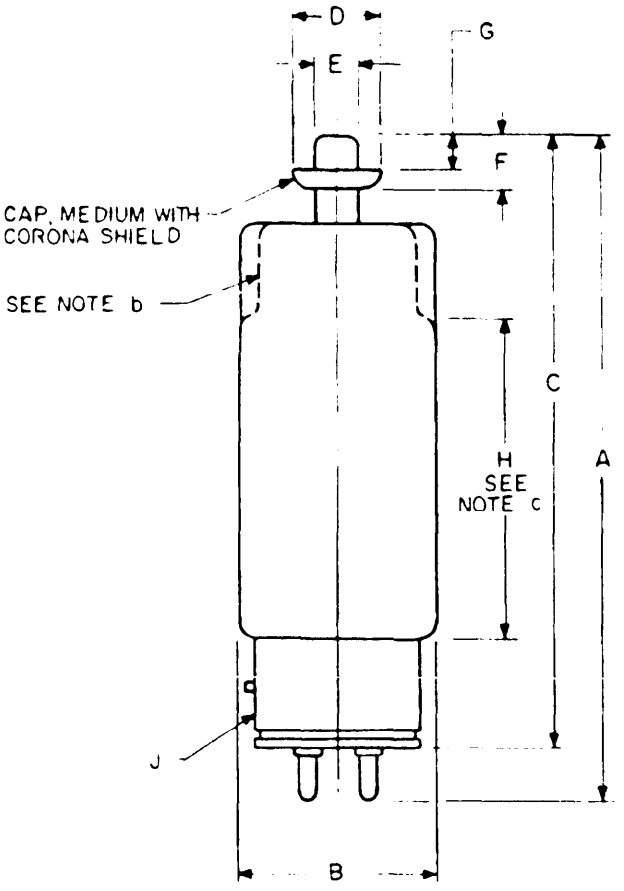
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User activities:
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Preparing activity:
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PIN NO	ELEMENT
1	g
2	h, k
3	h
4	k
CAP	a

LTR	DIMENSIONS IN INCHES WITH METRIC EQUIVALENTS (MM) IN PARENTHESES (SEE NOTE a)	
	MINIMUM	MAXIMUM
QUALITY CONFORMANCE INSPECT, PART 2		
A	8.250 (209.55)	8.750 (222.25)
B	2.313 (58.75)	2.563 (65.10)
C	7.531 (191.29)	8.031 (203.99)
QUALITY CONFORMANCE INSPECT, PART 3 (SEE NOTE d.)		
D	.994 (25.25)	1.170 (29.72)
E	.559 (14.20)	.573 (14.55)
F	.609 (15.47)	.641 (16.28)
G	.375 (9.53)	
H	4.250 (107.95)	
J	BASE: A4 - 18	

NOTES:

a. METRIC EQUIVALENTS (TO THE NEAREST .01 MM) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH=25.4 MM.

b. ALTERNATE SEAL CONFIGURATION.

c. CLAMPING SHALL BE CONFINED TO THE BASE AND THE AREA DESIGNATED BY DIMENSION H.

d. SEE NOTE 11.

FIGURE 1. Outline drawing of electron tube type 5C22

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